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Abstract

Retrieval-Augmented Generation (RAG) is an important way to mitigate hallucination in large language models (LLMs) by integrating relevant external knowledge. With the advance of multimodal LLMs, retrieval augmentation now extends not only textual tasks but also multimodal ones. In this paper, we conduct a comprehensive survey of the most recent work on multimodal RAG (MM-RAG) in the sense that it has a full coverage of almost all the combinations of modalities as input and output, whereas existing survey papers typically focus on one or two modalities. Based on different input-output modality combinations, we present a taxonomy of MM-RAG methods which gives us a much clearer picture of their key technical components. Facilitated by such a taxonomy, we identify four essential stages of the workflow of MM-RAG, summarize common approaches to each stage, and discuss optimization strategies for each modality. To provide a holistic understanding and practical guidance for building MM-RAG systems, we also discuss training strategies and evaluation methods of MM-RAG. Finally, we discuss various MM-RAG applications and future directions. Updated information related to this survey can be found at <https://github.com/INTREBID/Awesome-MM-RAG>.

A Comprehensive Survey on Multimodal RAG: All Combinations of Modalities as Input and Output

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Retrieval-Augmented Generation (RAG) is an important way to mitigate hallucination in large language models (LLMs) by integrating relevant external knowledge. With the advance of multimodal LLMs, retrieval augmentation now extends not only textual tasks but also multimodal ones. In this paper, we conduct a *comprehensive* survey of the most recent work on multimodal RAG (MM-RAG) in the sense that it has a full coverage of almost all the combinations of modalities as input and output, whereas existing survey papers typically focus on one or two modalities. Based on different input-output modality combinations, we present a taxonomy of MM-RAG methods which gives us a much clearer picture of their key technical components. Facilitated by such a taxonomy, we identify four essential stages of the workflow of MM-RAG, summarize common approaches to each stage, and discuss optimization strategies for each modality. To provide a holistic understanding and practical guidance for building MM-RAG systems, we also discuss training strategies and evaluation methods of MM-RAG. Finally, we discuss various MM-RAG applications and future directions. Updated information related to this survey can be found at <https://github.com/INTREBID/Awesome-MM-RAG>.

1 INTRODUCTION

Large language models (LLMs) have developed rapidly in recent years. They excel in the understanding and generation of not only texts [1–3], but also other modalities such as image generation [4, 5], audio recognition and synthesis [6, 7], and video generation [8–10]. Retrieval-augmented generation (RAG) is a promising solution for reducing the hallucination of LLMs by retrieving external knowledge. It also supports domain-specific applications via local knowledge bases, protecting data security. Compared to knowledge distillation and fine-tuning, RAG exploits external information and integrates new knowledge in real-time, with no need for additional training and updates on model parameters.

As multimodal large language models (MLLMs) progress, the RAG paradigm is extended beyond the single modality of texts to multiple modalities including images, audio, video, etc., giving rise to *multimodal* RAG (MM-RAG). The evolution from the traditional (textual) RAG to MM-RAG has gone through two phases. Initially, MM-RAG systems were limited to processing multimodal inputs and producing textual outputs. Subsequent advances in multimodal generative models enable MM-RAG systems to generate multimodal outputs. The input-output modality combination determines not

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only the architecture and capabilities of an MM-RAG system, but also the training strategies and evaluation methodologies employed during system development.

Several surveys [11–14] exist related to MM-RAG systems. However, they typically focus on one or two modalities of text and images, and none covers input-output combinations of various modalities such as text, images, audio, video, 3D data, and structured data. Zhao et al. [11] and Zhao et al. [12] explore diverse applications of MM-RAG in many modalities but do not systematically discuss its technical details, enhancement strategies, and evaluation approaches. Abootorabi et al. [13] focus on different modalities as input reflected by their paper’s title “Ask in Any Modality,” but pay little attention to multimodal output, let alone various combinations of modalities as input and output. Moreover, while they have a nice coverage of many MM-RAG papers, the analysis and discussions are rather limited due to the paper’s page limit. Mei et al. [14] decompose an MM-RAG system into five components: document parsing and indexing, search planning, retrieval, and generation, and give a nice survey of each component; however, most of the works surveyed are for the individual components in the generic setting (e.g., document parsing and retrieval in general), rather than systematically in the context of MM-RAG. In comparison, most of the works our survey covers are methods specifically designed for MM-RAG (e.g. pre-retrieval and retrieval in the context of MM-RAG).

This paper makes the first attempt to survey MM-RAG of all possible combinations of commonly used modalities as input and output, and introduces a taxonomy of MM-RAG methods based on the modality combinations. Specifically, as shown in Table 1, we cover studies that have text, image, code, audio, video, text+image, text+code, text+video, etc., as input and all these modalities as output (i.e., the generated content) as well. There is a total of 54 combinations in the table, and we have found existing work in only 18 of these combinations; the majority of the combinations remain unexplored and each may correspond to an important and exciting group of applications. Such a comprehensive coverage facilitates the understanding of various systems with respect to their key components, e.g., what model structures are typically used for taking text+image as input and generating text as output? It also provides inspiration on the vast, yet underexplored space in MM-RAG such as text+video as input and video as output. Our taxonomy offers a clear view on the key components determined by the input and output modalities of MM-RAG systems, and is very different from the taxonomy proposed by Abootorabi et al. [13], which is based on the different stages of MM-RAG systems.

Facilitated by such a taxonomy, we identify four essential stages of MM-RAG: *pre-retrieval*, *retrieval*, *augmentation*, and *generation*, summarize common approaches for each stage, and discuss optimization strategies for each input-output modality combination. To provide a holistic understanding and practical guidance for building MM-RAG systems, we also discuss training strategies and evaluation methods of MM-RAG.

The main contributions of this paper are summarized as follows:

- This is the first survey on MM-RAG that covers all possible combinations of commonly used modalities as input and output, and proposes a taxonomy from the perspective of modality combinations. Based on this taxonomy, we analyze existing studies and summarize key approaches for various modality combinations.
- We identify four essential stages of the workflow of MM-RAG and systematically analyze common approaches for each stage, and discuss modality-specific optimization strategies.
- We discuss training strategies and the evaluation of MM-RAG. Finally, we explore various MM-RAG applications and future work.

The paper is organized as follows. Section 2 provides the problem formulation of MM-RAG. Section 3 presents a taxonomy of MM-RAG methods based on input-output modality combinations and analyzes key approaches for each combination. Section 4 describes four essential stages of

MM-RAG. [Section 5](#) examines training strategies for MM-RAG systems and [Section 6](#) reviews the evaluation of MM-RAG. [Section 7](#) and [Section 8](#) discuss applications and future directions of MM-RAG. [Section 9](#) concludes this paper.

2 PROBLEM FORMULATION

An MM-RAG system typically consists of three fundamental components: a knowledge base, a retriever, and a generator. Below, we describe each component and then formulate the problem of MM-RAG. The paradigm of MM-RAG is shown in [Fig 1](#).

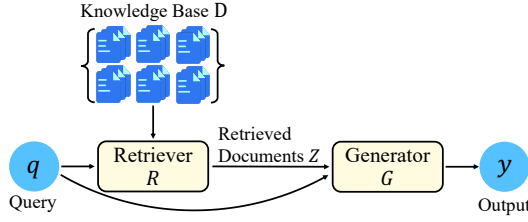


Fig. 1. Mathematical definition illustration of MM-RAG

The knowledge base D is a (typically large) collection of documents in the format of multiple modalities such as text, images, code, audio, videos, 3D data, and structured data, used to help the generator produce the output. Assume that an MM-RAG system operates in the semantic space V , which may include the text space V_T , the image space V_I , etc. An external knowledge base is formulated as $D = \{d_i\}_{i=1}^{|D|}$, where each document $d_i \in V$.

The retriever R retrieves the top- k relevant multimodal documents from D based on their relevance to a given user's multimodal query $q \in V$, and the retrieved documents Z is defined as:

$$Z = R(q, D) = \text{top-}k(\text{sim}(q, d)) = \{z_1, z_2, \dots, z_k\}$$

$d \in D$

where $\text{sim}(\cdot, \cdot)$ is a metric function to measure the relevance (e.g., similarity).

The generator G produces augmented outputs y based on the retrieved documents Z and the query q :

$$y = G(q, Z) \in V.$$

We formulate the problem of MM-RAG below.

Definition 1: *Multimodal Retrieval-Augmented Generation (MM-RAG).* Given a query q and a knowledge base D , which together has more than one modality, the goal of MM-RAG is to retrieve relevant documents Z from D and generate a response y based on q and Z such that the difference between y and the ground truth y^* is minimized under a given measure M :

$$\min_{R, G} M(y, y^*)$$

The ground truth is often not unique and difficult to obtain, so the evaluation of MM-RAG systems usually uses proxy measures such as retrieval quality metrics (e.g., Precision@k, Recall@k, nDCG) and generation quality metrics (e.g., BLEU, ROUGE for text and FID, IS for images). We elaborate on the evaluation of MM-RAG in [Section 6](#).

In this paper, a *document* refers broadly to any retrievable item in the knowledge base, including but not limited to text passages, images, or other modality-specific content.

3 TAXONOMY

The retriever and the generator are the two core components of an MM-RAG system, and the models to use as the retriever and the generator are determined largely by the input and output modalities. In this section, we present a taxonomy of MM-RAG methods according to the combinations of modalities as input and output, summarized in Table 1. For example, the input is a single modality of text and the output the multimodality of text+code; the input is the multimodality of text + image and the output the multimodality of text+image as well. A “✓” sign means that there is already work for that modality combination, and the majority of the cells being empty reveals plentiful unexplored opportunities. Based on this taxonomy, we summarize commonly used models and algorithms for the core components of MM-RAG in Table 2. Specifically, we discuss commonly used models for each modality as input in Section 3.1 and as output in Section 3.2.

Table 1. A Taxonomy Matrix for Existing MM-RAG Studies from an Input-Output Perspective

Input \ Output		Uni-Modal				Multimodal	
		Text (§ 3.2.1)	Code (§ 3.2.6)	Image (§ 3.2.2)/Audio (§ 3.2.3)/3D (§ 3.2.5)	Video* (§ 3.2.4)	Text+Image (§ 3.2.2)	Text+Code (§ 3.2.6)
Uni-Modal	Text (§ 3.1.1)	✓	✓	✓	✓		✓
	Image (§ 3.1.2)	✓					
	Code (§ 3.1.5)	✓	✓				✓
	Audio (§ 3.1.3)	✓					
	Video* (§ 3.1.4)	✓					
Multimodal	Text+Image (§ 3.1.2)	✓				✓	
	Text+Code (§ 3.1.5)	✓	✓				✓
	Text+Structured Data (§ 3.1.6)	✓					
	Text+Video (§ 3.1.4)	✓					

* Video may be considered a composite of audio and image frames. However, for analytical convenience, we treat it as a single modality in this study.

3.1 Input

For input, the key is to encode the query and knowledge base for retrieval such that they are aligned and can be later used for generation. Therefore, we focus on discussing the encoders for various modalities of data.

3.1.1 Text. Text is the most basic modality used for information retrieval. It is often used as a medium for associating data from other modalities to enable cross-modal retrieval. In MM-RAG, textual queries are typically encoded using text encoder-only models (e.g. BERT [75] and its variants [117–119]) or encoder-decoder models (e.g. T5 [17]). For example, RA-VQA [25] utilizes BERT to encode both textual queries and images through their captions for retrieval. Its successor [37] adopts ColBERTv2 [119] with a late interaction mechanism. MMed-RAG [38] uses a clinical domain BERT encoder, which is fine-tuned on specific clinical text for retrieval of medical documents.

Text is often combined with other modalities (e.g., image, video) to form a joint input in MM-RAG systems. Such multimodal inputs involve cross-modal retrieval and interaction, which require the encoder to align heterogeneous semantic spaces through additional training or fine-tuning. We will elaborate on such multimodal inputs in Section 3.1.2 to Section 3.1.6.

3.1.2 Image. When image is the only input modality and text is the output modality, the typical corresponding task is *image captioning*. Early MM-RAG studies with image input adopt traditional models such as CNNs and ResNet as encoders [44, 46, 47]. Recent MM-RAG studies with image input change to Transformer-based visual encoders such as ViT [120] and Swin-Transformer [121], which leverage attention mechanisms to enhance global feature modeling [26, 39, 122].

Besides being used alone, images are often combined with text to form multimodal input, typically in *visual question answering* (VQA) tasks [123–126]. The joint embedding of text and image inputs usually comes from a text encoder and an image encoder that process each modality independently. This results in unaligned semantic spaces and necessitates additional training to establish cross-modal alignment. For example, XTRA [29] and Maria [23] both use BERT and ResNet as encoders and then align the text and image embedding spaces with contrastive learning. MuRAG [24] and

Table 2. Core Components, Tasks, and Applications of MM-RAG Methods under different Input-Output Modality Combinations

Modality	Backbone		Task	Examples of MM-RAG systems
	Retriever	Generator		
T+I→T	Dual-Encoder Architecture Models(CLIP [15], BERT + ResNet, ViT + T5, etc.)	1. General LLM (GPT [16], T5 [17], UniLM [18], LLaMA [19]); 2. Visual LLM (GPT-4o [20], InstructBLIP [21], LLaVA [5])	Visual Dialog	KIF-Transformer [22], Maria [23], etc.
I→T			Visual QA	MuRAG [24], RA-VQA [25], RAMM [26], KAT [27], PICa [28], XTRA [29], RagVL [30], RACC [31], ReVeal [32], VISA [33], M3DocRAG [34], EchoSight [35], VisRAG [36], RA-VQAv2 [37], MMed-RAG [38], etc.
		Image Captioning	Re-ViLM [39], CRSR [40], SmallCap [41], RA-CM3 [42], RA-Transformer [43], Fei [44], EXTRA [45], DeltaNet [46], RAMP [47], FactMM-RAG [48], MuRAG [24], Zhan et al. [49], etc.	
T→I		1. Image-Specialized Generator (Diffsion Model, GAN); 2. Visual LLM (LLaVA [5], LaViT [50])	Image Generation	RetrieveGAN [51], FineRAG [52], ImageRAG [53], X&Fuse [54], TIGeR [55], Re-Imagen [56], RDM [57], KNN-Diffusion [58], Li et al. [59], etc.
A→T	1. Dual-Encoder Architecture Models (CLAP [60]); 2. Audio-Specialized Encoder (Wav2vec [61], VGGish [62], HuBERT [63])	General LLM (GPT [16, 64], T5 [17], UniLM [18])	Audio Captioning	RECAP [65], Koizumi et al. [66], etc.
T→A	Dual-Encoder Architecture Models (CLAP [60], RoBERT + HTS-AT)	1.Audio-Specialized Generator (HiFi-GAN [67], Diffusion Model, VAE); 2. Audio LLM (Whisper [6], QwenAudio [68])	Audio Generation	Make-An-Audio [69], Re-AudioLDM [70], Audiobox TTA-RAG [71], Xue et al. [72], Choi et al. [73], etc.
V→T	VideoMAE [74], CLIP [15]	1. General LLM (GPT [1, 16], BERT [75], LLaMA [19]); 2. Visual LLM (LLaVA [5], Video-LLaVA [76])	Video Captioning	EgoInstructor [77], KaVD [78], etc.
T+V→T			Video QA	VITA [79], R2A [80], iRAG [81], R-ConvED [82], VideoRAG [83], Video-RAG [84], VideoRAG [85], etc.
T→V		Diffusion Model	Video Generation	Animate-a-Story [86], etc.
T→3D	Dual-Encoder Architecture Models (CLIP [15])	Diffusion Model	3D Generation	ReDream [87], Remodiffuse [88], etc.
C→T	1. Sparse Retriever (BM25, AST, Edit Distance, Jaccard Index); 2. General LM (mT5 [89], SROberta [90]); 3. Code-Specialized Encoder (CodeT5 [91], GraphCodeBERT [92]);	1. General LLM (ChatGPT); 2. Code-Specialized LLM (CodeGPT-adapted [93], CodeT5 [91], CodeX [94], CodeGen [95])	Code Summa- rization	Rencos [96], RedCoder [97], HGNN [98], RACE [99], etc.
T→C			Code Generation	CEDAR [100], RepoCoder [101], RedCoder [97], Cocomic [102], EvoR [103], Synchromesh [104], Retrieve-and-Edit [105], ReAcc [106], DocPrompt [107], etc.
C→C			Text-to-SQL	XRICL [108], DUREPA [109]
			Code Completion	RepoCoder [101], ReACC [106], Cocomic [102], etc.
T+S→T	1. Dense Retriever (BERT [75], T5 [17]); 2. Query Language (SQL, SPARQL, Cypher);	General LLM (GPT [1, 16], BERT [75], LLaMA [19]);	KBQA	KnowledGPT [110], G-Retriever [111] ReTraCk [112], KAG [113], etc.
			Table QA	LI-RAGE [114], T-RAG [115], GTR [116], etc.

ReVeal [32] both combine pre-trained T5 and ViT models as encoders for textual and visual inputs, respectively.

Vision-language models (VLMs), a kind of MLLMs pre-trained on extensive text-image datasets, such as CLIP [15] and BLIP [127], achieve end-to-end text-image alignment, thus saving the effort of post-alignment [27, 42, 128]. For example, PNG-VQA [128] leverages BLIP to match image patches with textual questions. For visually rich documents containing text, images, tables, and complex layouts, some VLMs such as ColPali [129] produce multi-vector embeddings directly from full-page screenshots, meaning that all the elements in a document are treated as a unified image input.

3.1.3 Audio. When audio is the only input modality and text is the output modality, the typical corresponding task is *audio captioning*. Audio is usually regarded as a continuous time-series signal represented as waveforms or spectrograms. Audio-based RAG typically employs pre-trained audio embedding models as encoders, such as Wav2vec [61], VGGish [62] and HuBERT [63]. Large Audio-Language Models, such as CLAP [60] and Qwen-Audio [68], have also been widely adopted to process text and audio input [65, 72, 130, 131].

3.1.4 Video. Video offers high information density, enabling it to convey events, processes, and contextual details with greater clarity and detail than other modalities. It can also be considered a composite of multiple modalities such as texts, images, and audio. When video is the only input modality and text is the output modality, the typical corresponding task is *video captioning*. When video is combined with text as input, a typical task is *video question answering* (Video QA), where the text provides explicit queries or additional context to guide the reasoning over video content. *Large video language models* (LVLMs) are used to encode and understand input videos [79, 83, 132]. For example, EgoInstructor [132] leverages VideoMAE [74] and UniFormerV2 [133] to encode egocentric video clips and exocentric video clips, respectively. VideoRAG [83] employs InternVideo2 [134] to extract embeddings for input videos and text.

However, LVLMs incur substantial latency and resource consumption. A more preferred approach is to convert videos into textual descriptions, which usually involves technologies such as image captioning [83], object detection [81, 84], Optical Character Recognition (OCR) [84], Automatic Speech Recognition (ASR) [83, 84], etc., for subsequent processing.

3.1.5 Code. Code consists of characters and can thus be treated as a special form of text. Pre-trained language models may be utilized to capture its semantics [100, 108]. For example, CEDAR [100] uses SROBERTa [90] to encode code snippets as vectors, while XRICL [108] trains a Cross-lingual Exemplar Retriever based on mT5 [89] for text-to-SQL tasks. However, code exhibits stricter syntactic constraints and structures than natural language, necessitating syntactic parsing and symbolic logic modeling to capture its structured semantics. To address this, when modeling code as a distinct modality, specialized encoders are employed to transform code into an embedding space aligned with other modalities, primarily text. For example, RACE [99] and REACC [106] utilize CodeT5 [91] and GraphCodeBERT [92].

3.1.6 Structured Data. Structured data, mostly knowledge graphs and tables, has predefined formats and explicit semantic relationships, making it well-suited to serve as knowledge bases for MM-RAG. Such data is often combined with textual queries, enabling tasks like *knowledge-based question answering* (KBQA) and *table question answering* (Table QA).

Knowledge graphs' structure enables multi-hop reasoning paths and sub-graph retrieval based on entities extracted from queries, thereby achieving deep parsing of semantic relationships, even on large graphs with the advance of graph learning [135, 136]. For example, KnowledGPT [110] prompts LLMs to generate search language for knowledge base operations. ReTraCk [112] uses an Entity Linker and a Schema Retriever to generate SPARQL programs.

For scenarios with small-scale tables as input, T-RAG [115] flattens the tables into text and directly merges them with the query to construct prompts for LLMs. For scenarios with large-scale

complex tables or multi-table associations as input, DUREPA [109] can automatically generate intermediate SQL for further retrieval. GTR [116] transforms large-scale table corpus into a graph and performs a multi-stage granularity retrieval process to identify relevant tables.

3.1.7 Others. Other modalities of data, although less frequently seen, may be used as input. One example is time series, and the task is to forecast the subsequent values of the time series. While traditional methods rely solely on learning temporal dependencies from the input window itself, MM-RAG augments by retrieving similar time series from a knowledge base [137–139]. In this paradigm, the model’s input typically consists of a time series sequence and retrieved similar time series segments, such as climate indicators observed in the past few days. For small-scale single-channel time series data, the input time series query can be directly concatenated with retrieved similar historical segments as input to the predictor, such as in RAF [137]. For multi-channel complex time series scenarios, pre-trained temporal encoders are employed to extract high-dimensional features from dedicated knowledge bases for retrieval, such as in TS-RAG [138]. Additionally, TimeRAG [139] leverages Dynamic Time Warping (DTW) as a similarity measurement method, which optimizes cross-scale sequence retrieval accuracy through an elastic alignment strategy. By integrating retrieval into time series forecasting, RAG-based approaches effectively leverage historical analogies, alleviating domain shift and improving generalization.

3.2 Output

For output, the key is how to integrate the query with the retrieved documents to generate augmented content and what generators to use for various output modalities.

3.2.1 Text. MM-RAG provides significant enhancements for various modality-to-text generation tasks, such as image-to-text (*image captioning*), audio-to-text (*audio captioning*), video-to-text (*video captioning*), text+image-to-text (VQA and *visual dialog*), etc. While text generation capabilities are well-established in traditional RAG systems, MM-RAG raises new challenges. In MM-RAG scenarios, generators must understand and integrate heterogeneous multimodal inputs retrieved from cross-modal sources to produce coherent textual outputs.

One approach adapts language models to understand multimodal embeddings by exposing them to multimodal inputs. For example, ReCap [65] and Koizumi et al. [66] both utilize GPT-2 [64] for audio captioning, where the model takes the raw audio along with retrieved audio captions and generates a descriptive caption for the original audio as output. Maria [23] leverages UniLM [18], a unified pre-trained language model, to take a dialog context and retrieved images as input and generate responses in visual conversations. Re-ViLM [39] uses a pre-trained retrieval-augmented language model, RETRO [140], and further fine-tunes it to produce descriptions for an input image with retrieved similar captions.

Recent advances have focused on MLLMs (e.g., LLaVA [5], GPT-4o [1]), which are inherently equipped with both multimodal understanding and text generation abilities. Such models can directly leverage their multimodal understanding abilities for downstream tasks, typically requiring only lightweight adaptation rather than resource-intensive retraining on specialized multimodal datasets. RA-VQAv2 [37] and RACC [31] use BLIP2-Flan-T5-XL [141] and InstructBLIP-FlanT5-XL [21] for VQA. VideoRAG [83] and RAG-Driver [142] use Video-LLaVA [76] for video captioning. VisRAG [36] uses MiniCPM-V [143] and GPT-4o for QA on multimodal documents.

3.2.2 Image. RAG-based image generation aims at producing images by integrating retrieved multimodal information to enhance accuracy and control, improving on text-to-image tasks. For instance, when asked to generate the rare bird “Kakapo,” retrieved images and descriptions provide authentic features—such as its moss-green feathers and nocturnal traits—that base models often miss.

Typical frameworks for image generation include generative adversarial networks (GANs) [51, 59], diffusion models [54, 56–58], and their variations. For example, Li et al. [59] propose a multi-stage GAN-based architecture that generates images by leveraging retrieved image features in conjunction with a given text prompt. RetrieveGAN [51] adopts PasteGAN [144] to synthesize images according to retrieved image patches. KNN-Diffusion [58] employs a diffusion-based framework that leverages retrieved images to perform text-driven image editing while maintaining the appearance and characteristics of the original objects. Different from the methods that retrieve images, RDM [57], Re-Imagen [56], and X&Fuse [54] leverage diffusion-based models to generate images based on retrieved text.

Recently, VLMs have emerged as pivotal tools in image generation. For example, TIGeR [55] unifies text-to-image retrieval and generation via LaViT [50] and Seed-LLaMA [145]. RA-CM3 [42] leverages a Transformer-based LLM, CM3 [146], which can generate both text and images.

3.2.3 Audio. Most studies focus on text-to-audio tasks, where retrieved information (audio–text pairs or audio clips) is used to condition the audio generation. Two main paradigms have emerged: traditional pipeline-based methods using intermediate acoustic embeddings and newer approaches leveraging large audio models for end-to-end generation.

Some approaches that rely on intermediate representations use spectral generation models—such as diffusion models or variational autoencoders (VAEs)—to synthesize acoustic features like mel-spectrograms. These features are then converted into waveforms by neural vocoders, such as HiFiGAN [67], which transform spectral representations into time-domain audio signals for reconstruction. Re-AudioLDM [70] and Make-an-Audio [69] both retrieve relevant audio–text pairs and combine them with the original prompt to condition a diffusion-based model to produce a mel-spectrogram that is then converted into a waveform using HiFiGAN. Choi et al. [73] further validate this approach. It retrieves relevant audio clips rather than audio–text pairs and reconstructs Foley sound waveforms from diffusion-learned features of these retrieved clips using HiFiGAN-based vocoding. However, these traditional approaches often require complex multi-stage training and specialized domain expertise for optimal performance.

The development of large audio models is pushing audio generation technology into a new stage, making RAG-based audio generation easier to train and deploy. WavRAG [131] pioneers RAG-driven audio synthesis by leveraging GPT-4o and Qwen-Audio [68] as core generators with retrieved audio, speech, or text as supplementary information. Xue et al. [72] implement RAG-based text-to-speech synthesis via GPT-SoVITS with text and speech prompts guiding the waveform generation process. Gonzales et al. [147] achieve retrieval-augmented music generation using MusicGen [148] guided by augmented prompts. These advances demonstrate that large audio models streamline RAG-based audio generation through unified multimodal processing architectures.

3.2.4 Video. Video generation is an emerging field and is still in its early stages, largely constrained by the inherent spatio-temporal complexity of video data and the substantial computational demands involved in generating high-fidelity content. These challenges make both video modeling and scaling extremely difficult. Therefore, the integration of MM-RAG into video generation is an even newer direction. Efforts such as Animate-A-Story [86] have made a first step. It extracts depth maps of retrieved video clips as motion structures and conditions a diffusion-based generator on these structures to synthesize coherent scenes. However, such attempts remain limited in scope, largely confined to basic temporal dynamics and elementary narrative structures.

3.2.5 3D data. RAG-based 3D data generation is in its nascent stage, with diffusion models serving as the implementation backbone. It is primarily applied in text-to-3D tasks. Current methods often follow a multi-stage pipeline: first providing generation guidance through 2D prior information,

followed by 3D data generation using models such as Stable Diffusion [149] or Score Distillation Sampling (SDS) [150]. For example, ReDream [87] employs an SDS-driven framework to generate 3D data with retrieved 3D assets as references for 3D appearances and geometries. ReMoDiffuse [88] uses retrieved 3D motion samples to guide the denoising process to support 3D motion generation via diffusion models combined with multi-level Transformer decoders.

3.2.6 Code. The generator integrates retrieved relevant artifacts—such as API documentation, cross-file code snippets, issue reports, and runtime traces—to produce more accurate and context-aware code, enabling tasks like *code generation* [101, 107] and *code completion* [102, 106]. General-purpose LLMs, by virtue of being exposed to corpora that contain substantial amounts of code, inherently acquire a certain degree of code generation capability. Nevertheless, code generation still requires strict syntactic constraints and logical dependencies. Therefore, domain-specific code models such as CodeX [94] and CodeGen [95], which undergo specialized optimization and fine-tuning on code-related data, achieve better performance in software engineering and related applications.

3.3 Insights from the Taxonomy

Our taxonomy gives a comprehensive overview of MM-RAG methods with all the common modality combinations as input and output, which provides insight on the design of new MM-RAG methods. Specifically, empty cells in [Table 1](#) represent input-output modality combinations that have never been explored and hence suggest novel MM-RAG applications and methods. For example, the empty cell with image as input and code as output implies an application of image-to-code RAG, such as generating the Scalable Vector Graphic (SVG) code from a given image as the query augmented by the retrieval of similar SVG codes. As another example, the cell with text+image as input and an image as output corresponds to the application of text+image-to-image RAG, which may generate more refined or stylistically specific images from textual prompts and a given image as the query augmented by composed image retrieval [151].

Based on the taxonomy, [Table 2](#) summarizes commonly used models and algorithms for the core components of MM-RAG (retrievers, generators, processing strategies, etc.) per input-output modality combination. For example, for text+image-to-text and image-to-text, commonly used retrievers are dual-encoder architecture models such as CLIP [15], BERT+ViT, etc., and the commonly used generators are general LLM or visual LLM. [Table 2](#) gives great guidance for designing new MM-RAG methods.

4 WORKFLOW

The previous section established a taxonomy of MM-RAG methods. Our taxonomy is based on input-output modality combinations, which mostly concern two core components, the retriever and the generator. For a functional MM-RAG system, we identify four essential stages of its workflow: *pre-retrieval*, *retrieval*, *augmentation* and *generation*, as shown in [Figure 3](#). *Retrieval* and *generation* involve the retriever and generator, respectively. *Pre-retrieval* involves knowledge base and query preparation. *Augmentation* involves the preprocessing of the query and retrieved documents before they are fed into the generator. For each stage, we discuss common approaches and modality-specific optimization strategies, and summarize representative studies in [Figure 2](#).

4.1 Pre-Retrieval

The pre-retrieval stage performs *knowledge base preparation*, which constructs the multimodal knowledge base to be retrieved and *query preparation*, which pre-processes users' queries to facilitate effective retrieval.

4.1.1 Knowledge Base Preparation. The raw data from external sources usually goes through preprocessing before being stored in a knowledge base following a certain organization format.

Data Preprocessing usually involves cleaning, chunking, and metadata attachment in traditional text-only RAG. In MM-RAG, these operations are adapted to process heterogeneous modalities into forms that are easy to process, such as text and images. For instance, ViTA [79] and iRAG [81] utilize LVLMS to generate text descriptions of videos. Redream [87] converts 3D data into images, which are then used as search keys for retrieval. VideoRAG [83] treats videos as frames for retrieval.

Data Chunking is a well-explored topic in RAG research. Off-the-shelf chunking approaches for text, such as fixed-size, sliding window, and recursive chunking, remain applicable to MM-RAG [152]. Images may be split into regions of interest (ROIs) via object detection, refined recursively or segmented with sliding windows. Audio, video, and text all have sequence structure, allowing similar chunking methods. Code chunking is easy due to its inherent syntactic structure.

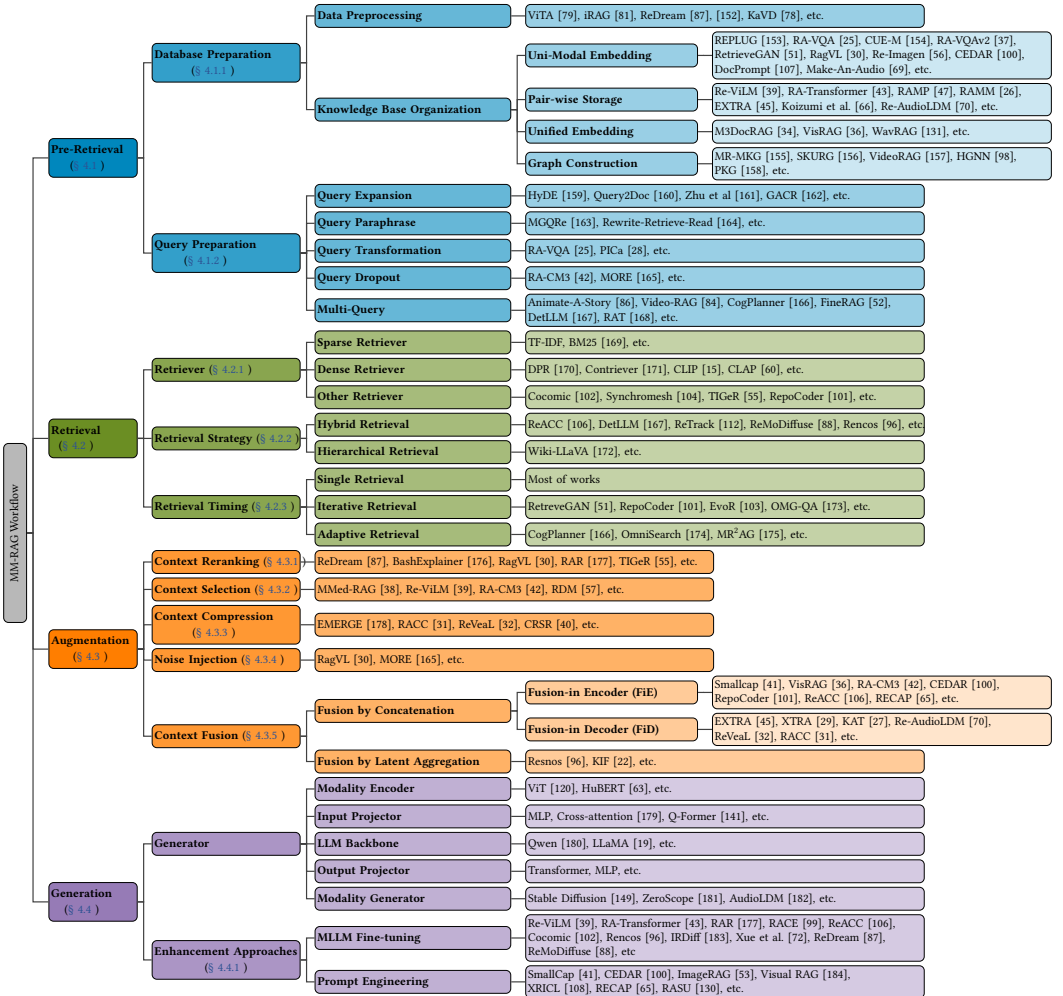


Fig. 2. MM-RAG Workflow.

Attaching structured metadata to multimodal content helps quickly filter and retrieve relevant documents. For instance, leveraging the metadata (tags and dates) attached to videos, KaVD [78] filters and retrieves time-sensitive news articles to ground the generation of video descriptions.

Knowledge Base Organization concerns how data is stored and organized in the knowledge base, which greatly affects the efficiency of retrieval. In traditional text-only RAG, the documents in the knowledge base are converted into embeddings through a text encoder such as BERT and then indexed in a vector database.

In MM-RAG, although some organization principles (e.g., embedding-based retrieval) are inherited, they are extended to accommodate multiple modalities, necessitating specialized encoders, cross-modal alignment strategies, and more complex storage patterns. There are four approaches: Uni-modal Embedding, Pair-wise Storage, Unified Embedding, and Graph Construction.

Uni-modal Embedding encodes multimodal data into separate embedding spaces through modality-specific encoders and stores them in separate databases indexed by each modality. During retrieval, a multimodal query is initially decomposed into modality-specific sub-queries (e.g., decomposing a query into textual and visual parts). Each sub-query then retrieves relevant results from the database corresponding to its modality. This approach builds on the simplicity and proven effectiveness of uni-modal RAG methods.

However, the decomposition step may damage the cross-modal semantic relationships, thereby limiting its applicability to scenarios with loose inter-modal correlations. For example, given a photo of a rare bird "Kakapo" and the question "What is unique about the bird's behavior?", uni-modal embedding may retrieve visually similar birds from the image database but fail to connect them with textual knowledge about its behavior.

Pair-wise Storage leverages paired data between two modalities, such as image-text pairs in image captioning datasets. During retrieval, the keys of the same modality as the query are searched through and the values corresponding to the most similar keys are returned [26, 39, 43, 47, 66, 70]. For example, Re-ViLM [39] constructs its knowledge base using a dataset consisting of image-text pairs, organized as a key-value map in which image embeddings serve as keys and the associated image captions serve as values. Likewise, Koizumi et al. [66] and Re-AudioLDM [70] build audio-text pairs for text-to-audio generation.

This pair-wise storage approach relies on a large amount of pre-constructed paired multimodal data. However, such datasets are scarcely available, which restricts the method's practical deployment.

Unified Embedding utilizes an embedding model that aligns multimodal data into one semantic space. The embedding vectors are then directly stored in the knowledge base as keys for retrieval. For example, M3DocRAG [34] and VisRAG [36] both treat documents as images and leverage VLMs (e.g., ColPali [129]) to encode them into a unified multimodal embedding space, thereby preserving full original information and avoiding data loss typically incurred during parsing. WavRAG [131] adapts Qwen2-Audio [68] into a powerful multimodal encoder for the knowledge base whose instances may be audio, text, or a combination of both. There have been attempts to align more than two modalities [185–187], but the capability of these models is still quite limited, and therefore their applications in MM-RAG are limited as well.

Graph Construction refers to organizing multimodal documents into graph-structured data so that they can then be retrieved using graph-based methods. Most notably, this method was first introduced in GraphRAG [188], but it incurred high computational costs and slow responses; LightRAG [189] introduced a lighter and simpler RAG framework, demonstrating the great potential of this class of methods. The powerful capabilities of LLMs such as entity extraction and relation linking make it easier to convert unstructured multimodal documents into structured

graph representations. This approach is suitable for scenarios that require reasoning and multi-hop question answering.

Notably, the feasibility of graph structuring depends on whether the modality contains explicit or derivable discrete units and relationships. For images, objects in images and their spatial relationships provide a basis for graph construction. MR-MKG [155] transforms the scene graph, which describes the objects, their attributes, and relationships in an image, into a multimodal knowledge graph. SKUGR [156] uses a NER model and a relation extraction model to generate graphs from textual and image inputs, thereby producing a more semantically rich and contextually relevant representation. For videos, discrete units such as frames or detected objects can be connected through temporal or spatial relations. VideoRAG [157] transforms videos into structured textual knowledge through image-to-text and audio-to-text conversion for enhanced indexing and retrieval. For code, functions, variables, and dependency calls naturally form nodes and edges. HGNN [98] represents source code as a *code property graph* and performs retrieval on it for summarization. PKG [158] generates a *programming knowledge graph* from a given dataset containing text and code content, where each node represents a code block extracted from a function’s context-flow graph. However, better methods are to be developed to build graphs for modalities like audio and 3D data.

4.1.2 Query Preparation. The original query may be ambiguous, incomplete, or misaligned and needs to be refined. This section covers five key techniques for MM-RAG query preparation: *query expansion*, *query rewriting*, *query transformation*, *query dropout*, and *multi-query*.

In MM-RAG, these techniques need to be adapted to heterogeneous modalities. This involves not only adapting text-oriented strategies to non-text inputs but also addressing modality-specific challenges such as cross-modal alignment, transformation, and semantic preservation for high-dimensional continuous embeddings.

Query Expansion enhances the original query through semantic expansion, logical refinement, or adding context to make the query more precise and help the system retrieve relevant documents effectively. HyDE [159] and Query2Doc [160] employ LLMs to generate a potential answer or pseudo document based on the original query, which typically shares significant semantic relevance with the target answer. It then concatenates the original query with the generated output to construct an expanded query. Zhu et al. [161] leverage a VLM during multi-round retrieval interactions to generate captions for retrieved images and append them to the original query, thereby achieving query expansion. GACR [162] employs an LLM to synthesize code snippets from natural language documentation to expand the query for retrieval. Text-based queries, including text-based code queries which possess explicit semantic information, are the most straightforward to expand. However, there has been limited research on expanding queries in other modalities for RAG.

Query Rewriting paraphrases the query via different expressions and information perspectives while retaining the semantics to broaden the coverage of various possible interpretations.

Textual query rewriting has been extensively explored, allowing controllable paraphrasing while preserving semantic fidelity. For example, MGQRe [163] leverages an MLLM to capture the retriever’s expression preferences and designs a three-stage reinforcement learning framework that guides an LLM-based rewriter to learn these preferences, ensuring that the rewritten queries are well aligned with the retriever’s preferences. Rewrite-Retrieve-Read [164] employs T5 as a rewriter, whose training relies on reinforcement learning guided by feedback provided from the generator.

For non-textual queries such as images, audio, or video, it is difficult to ensure both semantic preservation and useful variation; therefore, rewriting them remains underexplored.

Query Transformation achieves cross-modal alignment by converting one modality (e.g., images) into another (e.g., textual descriptions) using expert models. For example, RA-VQA [25] and PICa [28] leverage OCR, object detection, and image captioning to transform the visual information

of a query into texts and combine with the textual part of the query for subsequent retrieval. Its primary advantage lies in simplifying multimodal interactions, enabling the reuse of off-the-shelf text-based RAG methods, and reducing computational costs.

However, the quality of transformation heavily depends on the representational capacity of the expert model. Critical semantic loss during the transformation or the introduction of noise may lead to retrieval or generation bias. Furthermore, intricate interdependencies between modalities are difficult to fully preserve through unidirectional transformation.

Query Dropout [42] randomly masks part of the original query to generate queries composed of its simplified forms so that different aspects or local parts of the query are attended, which increases the diversity of retrieved documents. To enhance the utilization of retrieved documents, MORE [165] introduces a new query dropout strategy during training, which drops the entire query so that the model is guided to rely more on retrieved evidence when producing answers, instead of depending only on its internal parameters.

The above textual query dropout approaches may be applied to other modalities. For example, certain regions of an image query can be masked to vary the focus on different visual features; for audio, segments of the waveform or spectrogram may be dropped to highlight alternative temporal or frequency patterns.

Multi-Query converts complex queries into multiple sub-queries or perspectives to increase both the breadth and depth of retrieved documents.

Query Split constructs multi-query by directly truncating the query. Text queries may be truncated at the sentence, phrase, or keyword level. In *Animate-A-Story* [86], the query, which is a story script for video synthesis, is split into several individual plots through storyboard analysis. Image query split may divide an image into multiple patches; audio query split may perform time or frequency domain segmentation. Video-RAG [84] splits the query via LVLM and generates retrieval requests for three distinct databases: OCR database, ASR database, and Object Detection database.

Query Decomposition decomposes a complex query into multiple sub-queries, each targeting a specific dimension of the original query, thereby enabling the retriever to capture relevant information from diverse perspectives [52, 166].

Chain-of-Query breaks a complex query down to a sequence of logically linked sub-queries to guide the retrieval process progressively leveraging the Chain-of-Thought (CoT) approach. For example, DetLLM [167] transforms a complex multi-hop question into a sequence of simpler single-hop queries by engaging in multiple interactions with an LLM. RAT [168] begins with initial reasoning steps generated by an LLM in a zero-shot setting and then iteratively refines each step utilizing the retrieved information. While both query decomposition and chain-of-query aim to handle complex queries, the former emphasizes breadth coverage, whereas the latter focuses on deep reasoning.

4.2 Retrieval

The quality of retrieval greatly influences the accuracy and diversity of the generated output. This section explores different types of retrievers, retrieval strategies, and the timing of retrieval.

4.2.1 Retriever. Retrievers perform the following steps: (1) encode multimodal documents into vector representations, i.e. embeddings; (2) build indexes on these representations; (3) given a query, retrieve relevant data objects such as entities or paragraphs in documents, which are then used by the generator.

In MM-RAG, retrievers support cross-modal tasks, which allow a query of one modality to retrieve documents of another, while retrievers in traditional text-based RAG are limited to retrieving text for text queries. Two common approaches to retrieval are sparse and dense retrievers.

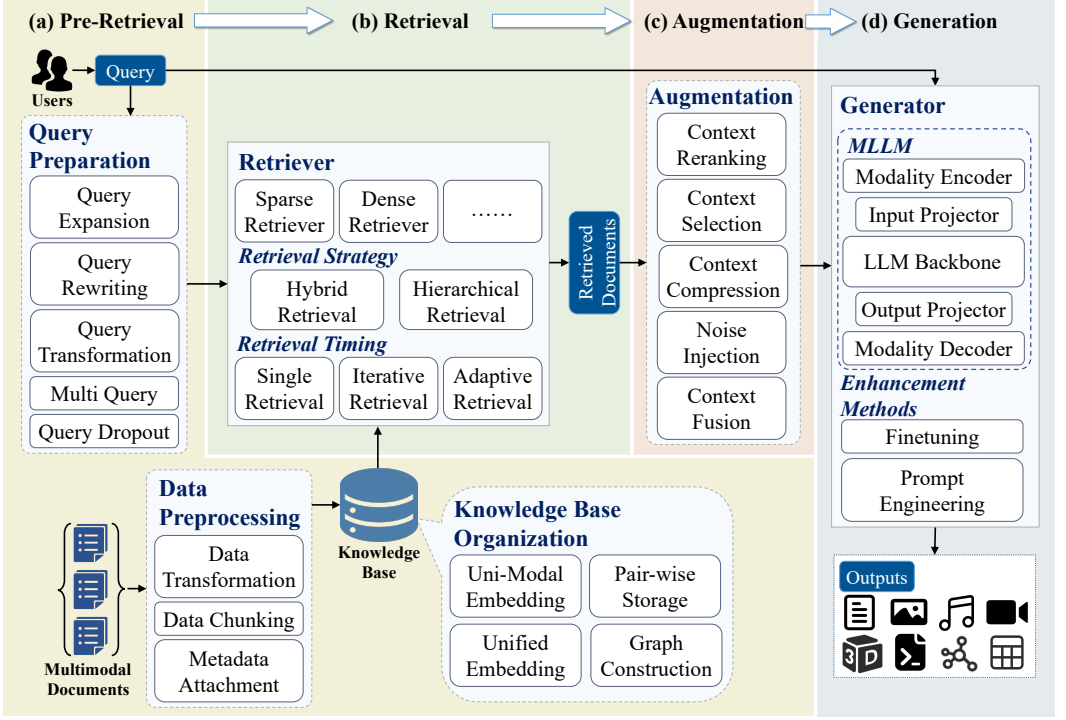


Fig. 3. The workflow of MM-RAG

Sparse Retrievers evaluate how relevant a query is to documents by applying statistical techniques based on term frequency (TF) and inverse document frequency (IDF). Typical algorithms include TF-IDF and BM25 [169]. However, sparse retrievers are constrained to keyword matching within a fixed vocabulary, lacking a deep semantic understanding of the content, so they only work for text (including code). For images and audio, they do not have the concept of “words,” and therefore keyword matching does not work. They usually require dense retrievers.

Dense Retrievers typically consist of a query encoder and a data encoder. After the raw data is encoded into dense embeddings and stored in vector databases such as Faiss [190], Milvus [191], Weaviate [192], and LanceDB [193], a query is also encoded into a query vector and searches for similar data object embeddings from vector databases using *approximate nearest neighbor* (ANN) algorithms such as locality-sensitive hashing [194] and HNSW [195]. Notable dense retrievers for text include DPR [170] and Contriever [171], and dense retrievers for other modalities include CLIP [15] and CLAP [60].

Other Retrievers are designed to cater to ad hoc use cases. For example, in some code-related tasks, retrievers perform retrieval based on the edit distance between code snippets or their corresponding abstract syntax trees (ASTs) [102, 104]. In TIGer [55], images from the knowledge base are tokenized into discrete codes and are maintained in a lookup table; retrieval is performed by forward beam search with a special MLLM. RepoCoder [101] calculates the similarity between code to be completed and candidate code snippets using Jaccard index.

4.2.2 Retrieval Strategy. We discuss representative retrieval strategies below.

Hybrid Retrieval integrates multiple retrievers, each may be optimized for a modality. For example, ReACC [106] adopts BM25 as a sparse retriever and GraphCodeBERT as a dense retriever and calculates a linear combination of their scores as the retrieval score for code retrieval. DetLLM [167] utilizes Contriever [171] and BM25 to retrieve unstructured knowledge while leveraging a SPARQL (a query language for structured data in knowledge bases) generator to access structured knowledge. ReTrack [112] employs a *named-entity recognition* (NER) system as an entity linker to identify explicit entities in the query and a dense retriever as a schema retriever to retrieve relevant types and relations mentioned either explicitly or implicitly in the query. ReMoDiffuse [88] incorporates both kinematic and semantic similarity measures to select the most informative data samples for effectively guiding the denoising process of a diffusion model for motion generation. Rencos [96] measures similarity on ASTs to retrieve syntactically similar code snippets and employs a Bi-LSTM-based dense retriever to identify semantically related code snippets.

Hierarchical Retrieval organizes documents into a layered structure, where higher-level nodes summarize or abstract the content of their subordinate nodes. This hierarchical organization facilitates a more structured and navigable retrieval process. For example, Wiki-LLaVA [172] employs a hierarchical retrieval process consisting of two stages. First, it retrieves the top- n most relevant documents. Then, within each selected document, it extracts the top- k most relevant passages. The resulting $n \times k$ text segments are fed into the generator. Most existing hierarchical retrieval methods have focused on text data (e.g. GraphRAG [188], HiRAG [196]) and rarely on data of other modalities.

4.2.3 Retrieval Timing. When and how many times to perform retrieval are important. We examine three typical choices of retrieval timing below.

Single Retrieval performs retrieval only once and is most common, but it may not provide thorough answers for complex queries.

Iterative Retrieval alternates between retrieval and generation across multiple iterations. During each iteration, subsequent retrieval is optimized based on all the content generated previously, progressively improving the response. For example, RetrieveGAN [51] applies this mechanism to select image patches aligned with earlier choices, while RepoCoder [101] and EvoR [103] use iterative retrieval-generation loops to reformulate queries and improve synthesis for code generation. OMG-QA [173] shows that multi-round retrieval outperforms single-round strategies in document recall in VQA tasks. While iterative retrieval captures the complexity of queries more effectively, it increases computational overhead and response time. Furthermore, it is difficult to determine the optimal number of iterations.

Adaptive Retrieval dynamically decides when to stop retrieval and proceed to generation. This strategy requires the model to autonomously adjust its retrieval behavior based on its understanding of the current context, striking a balance between generating accurate content and conserving response latency. For example, CogPlanner [166] employs an MLLM as the planning expert who analyzes the current multimodal content, evaluates the information gathered thus far, and subsequently determines the most appropriate follow-up retrieval action and reformulates the query. OmniSearch [174] proposes a self-routing framework with a planning agent, which comprehends the query and retrieval feedback and determines the follow-up sub-question to tackle. MR²AG [175] lets an LLM output special tokens first to distinguish whether the user's query can be answered without external knowledge.

4.3 Augmentation

Retrieved documents often contain irrelevant or redundant information. Additionally, the documents come from different modalities, and their heterogeneous formats and representations make

it difficult to process effectively by generators. The augmentation stage preprocesses them before passing them to the generator by one or several of the following techniques: *context reranking*, *context selection*, *context compression*, *noise injection*, and *context fusion*, detailed below.

4.3.1 Context Reranking. During retrieval, efficiency is often prioritized over precision, which may result in the inclusion of irrelevant or suboptimal candidates. A reranker addresses this by evaluating the retrieved documents using more sophisticated models or rules to assign more accurate relevance scores, which are used to rerank the documents. Text-only RAG typically uses language models, lexical similarity, or dense text embeddings.

In MM-RAG, context reranking incorporates multimodal signals from both textual and non-textual content, using alternative scoring functions for reranking. For example, ReDream [87] uses the joint embedding of text and images from CLIP for reranking. BashExplainer [176] uses text edit distance for lexical similarity reranking.

MLLMs are more commonly utilized to rerank multimodal documents, which either leverage their inherent capabilities or are fine-tuned for reranking through approaches such as instruction tuning [30, 177]. TIGeR [55] proposes a reverse re-ranking strategy, which re-ranks retrieved images by using an MLLM as an image captioner to calculate the probability of the textual query X given the retrieved image Y , $p(X|Y)$.

4.3.2 Context Selection. The goal of context selection is to optimize the retrieved documents by setting rules to eliminate irrelevant or redundant information. For instance, MMed-RAG [38] proposes an adaptive context selection strategy that truncates the retrieved documents when the relevance scores between retrieved documents sharply decline after a certain number of results.

Even for content that directly matches the query, selectively removing parts of it may prevent the model from overly relying on retrieved results for "copycat" answers, thereby enhancing the model's generative and generalization capabilities. Re-ViLM [39] removes the retrieved image-text pairs when one element of them is identical to the input query. RA-CM3 [42] excludes candidates that exhibit high similarity to the query or previously retrieved documents, thereby reducing redundancy.

Beyond filtering content that closely matches the query, retrieved documents that are highly similar to each other cause unnecessary computation. Therefore, some approaches retain only representative instances among similar documents to enhance efficiency [57].

4.3.3 Context Compression. Context compression compresses the retrieved documents to highlight their essential content. For textual content, it is often through summarization [178].

In MM-RAG, however, context compression involves reducing the dimensionality or size of embeddings and meanwhile preserves as much semantics as possible [31, 32, 40]. For example, RACC [31] compresses retrieved information by learning to generate compact, fixed-length soft prompts from the original documents using an MLLM. ReVeal [32] utilizes a perceiver [197] architecture to compress the retrieved documents and incorporates two supplementary regularizations to enhance the expressiveness of the resulting embeddings.

4.3.4 Noise Injection. Irrelevant, redundant, or misleading information, which we refer to as *noise*, exists in the retrieved documents. Traditional text-only RAG systems typically treat noise as a harmful factor and focus on its removal through retrieval refinement or filtering, thus the idea of deliberately *injecting* noise during training is far less common in traditional RAG.

In MM-RAG, noise injection emerges as a distinctive strategy to enhance model robustness to handle imperfect retrieved documents. For instance, RagVL [30] incorporates noise at both data level and token level during training to enhance the model's resilience and visual comprehension. MORE [165] injects noise by replacing retrieved documents with unrelated content from other

data samples, encouraging the model to discern useful information and avoid relying overly on retrieved results. Furthermore, studies have found that some types of noise [198] can enhance model performance [199, 200].

4.3.5 Context Fusion. Context fusion integrates the query with the retrieved documents to help understand the query, which falls into two categories: *fusion by concatenation* and *fusion by latent aggregation*.

Fusion by concatenation primarily takes in two forms:

Fusion-in-Encoder (FiE) utilizes fixed prompt templates (e.g., *Answer the following question according to the relevant documents and images. Question: {query}, Relevant Document: {doc}, Images: {image}, Answer:*) to form the input that is then fed into the generator. It is particularly suited to scenarios involving retrieved documents that are limited in size and offers easy implementation, widely adopted in MM-RAG methods [36, 41, 42, 65, 100, 101, 106]. This approach has several limitations. First, it exhibits the *lost-in-middle* problem [201] when processing extensive or multiple documents, i.e., the model’s capability of processing information in middle positions deteriorates. Second, the method is inherently constrained by the model’s context window, making it difficult to accommodate a large number of lengthy documents, which limits its applicability in knowledge-intensive scenarios.

Fusion-in-Decoder (FiD) [202] encodes each document independently, concatenates the resulting embeddings, and only performs fusion in the decoder, where the model jointly attends over all document embeddings during generation. Unlike FiE, which concatenates documents before encoding, FiD avoids the input length bottleneck of the encoder and enables the decoder to integrate more documents, thereby offering stronger scalability to larger retrieval sets. For scenarios where many retrieved multimodal documents need to be integrated, this approach may be adopted, where modality-specific encoders (e.g., text-only or image-only) produce separate embeddings that are fused in the decoder for cross-modal reasoning [27, 29, 45, 70]. However, the quadratic time complexity of the attention mechanisms results in rapidly increasing computational costs as context length grows, which limits the efficiency of FiD in very large-scale settings.

Recent MM-RAG methods incorporate retrieval scores into the fusion process, letting the decoder pay more attention to more relevant documents [31, 32]. For example, ReVeal [32] computes a soft attention mask based on retrieval scores as a prior, which is directly injected into the attention calculation to guide the fusion process.

Fusion by latent aggregation combines multiple document embeddings into a single context vector by assigning a weight to each document and computing a weighted sum, where the weights may be learned or manually defined. For example, Resnos [96] employs a manually tuned hyperparameter to perform a weighted summation of the retrieved document embeddings based on their relevance scores. KIF [22] introduces a gating mechanism to fuse the retrieved documents. Despite the fact that summing embeddings may cause information loss and semantic interference between documents, it provides a simple, scalable, and efficient way to integrate multi-document contexts.

4.4 Generation

The generation module, called the *generator*, generates the output given the input, including the queries, retrieved documents, and prompts. In traditional text-based RAG, the generator is usually an LLM and only receives and produces text outputs. In MM-RAG, the generator is usually an MLLM and supports multimodal inputs and outputs.

Previous studies [203, 204] have provided detailed explanations of the architecture of MLLMs. One [204] focuses on multimodal inputs and the other [203] focuses on multimodal outputs. Zhang

et al. [203] suggest that an MLLM consists of five components: "*Modality Encoder, Input Projector, LLM Backbone, Output Projector, and Modality Generator*", described as follows.

- *Modality Encoder* takes raw inputs of different modalities and encodes them into features. The features of different modalities require distinct encoding approaches and models, such as ViT [120] for image and video, and HuBERT [63] for audio. Different from encoders in dense retrievers that aim to obtain embeddings for retrieval, the modality encoder here is designed to capture modality-specific semantic representations, preserving rich information necessary for downstream multimodal understanding and generation.
- *Input Projector* (referred to as Connector in [204]) maps the encoded features of different modalities into a feature space that can be understood by the LLM Backbone, which primarily operates in a text vector space. A simple Input Projector can be implemented with an MLP to align modality gaps, or with more complex modules such as cross-attention [179] and Q-Former [141].
- *LLM Backbone* serves as the core part of the MLLM, typically accounting for about 80% of the model's parameters [204], and is responsible for processing embeddings from different modalities. It performs understanding, reasoning, and decision-making based on the input semantics. It directly generates text output or signals of other modalities if necessary. Examples include Qwen [180] and LLaMA [19], etc.

For non-text modalities, additional components are required to interpret and decode the intermediate embeddings produced by the LLM backbone into the desired modality using the following components:

- *Output Projector* maps the token embeddings from the LLM Backbone to a feature space that *Modality Generator* can understand. This is typically achieved using a Transformer or MLP with learnable decoder feature sequences.
- *Modality Generator* generates outputs in various modalities. Current studies mainly use latent diffusion models as generators, such as Stable Diffusion [149] for image generation, ZeroScope [181] for video generation, and AudioLDM [182] for audio generation.

However, there is still a lack of unified, fully multimodal large models that can seamlessly handle arbitrary combinations of input and output modalities within a single architecture. Developing such a unified multimodal model remains an open challenge.

4.4.1 Enhancement Methods. There are two common methods that can enhance the effectiveness of a generator: prompt engineering and MLLM fine-tuning.

Prompt Engineering freezes the model's parameters but guides the model to generate better output through carefully designed input prompts, thus offering flexibility and opportunity for improvement. However, manually crafting prompts is usually via trial-and-error and is time-consuming, limiting scalability and consistency. In MM-RAG, prompt engineering is primarily employed to integrate the query with the retrieved contents, as elaborated in [Section 4.3.5](#).

MLLM Fine-tuning adapts an MLLM to specific tasks either by enhancing domain-specific knowledge or by achieving stronger alignment with a retriever, as elaborated in [Section 5.2.1](#).

5 TRAINING STRATEGIES

Training helps models to acquire the capability to retrieve and generate multimodal content. There are mainly two strategies: *parameter-frozen strategies* ([Section 5.1](#)) and *parameter-trainable strategies* ([Section 5.2](#)). We summarize training methods in [Table 3](#) and commonly used training datasets in [Table 4](#) per input-output modality combination.

Table 3. Training Strategies of MM-RAG

Training Strategy		Methods
Parameter-Frozen Strategy		PICa [28], TiGeR [55], RAR [177], CEDAR [100], RepoCoder [101], EvoR [103], PKG [158], ViTA [79], VideoRAG [83], VideoRAG [157], Video-RAG [84], iRAG [81]
Parameter-Trainable Strategy	Modular Training	VisRAG [36], DocPrompt [107], Synchromesh [104], SmallCap [41], RACC [31], EchoSight [35], RECAP [65], MR-MKG [155], EMERGE [178], MORE [165], FactMM-RAG [48], RAMM [26], XTRA [29], RA-VQAv2 [37], MMed-RAG [38], Maria [23], RagVL [30], RACE [99], ReACC [106], HGNN [98], Rencos [96]
	End-to-End Training	MuRAG [24], RA-VQA [25], ReVeal [32], RetrieveGAN [51], RA-CM3 [42], RASU [130]

Table 4. Commonly Used Datasets for Different Modality Tasks

Modality	Datasets
Text ↔ Image	LAION [205], MS-COCO [206], Flickr30K [207], NoCaps [208], WIT [209]
Text+Image → Text	OK-VQA [125], A-OKVQA [126], WebQA [210], MultimodalQA [211], Encyclopedic-VQA [212], InfoSeek [213], ScienceQA [214], ArxivQA [215]
Text ↔ Audio	AudioCaps [216], AudioSet [217], CommonVoice [218], Clotho [219], MusicCaps [220]
Text ↔ Video	MSR-VTT [221], ActivityNet Captions [222], VaTeX [223], Ego4d [132], HowTo100M [224]
Text+Video → Text	MSRVTT-QA [225], MSVD-QA [225], NextQA [226], ActivityNet-QA [227], MLVU [228], TGIF-QA [229], KnowIT VQA [230], Video-MME [231], LongVideoBench [232]
Text ↔ 3D	Objaverse [233], HumanML3D [234], KIT [235]
Text ↔ Code	CodeXGLUE [93], CoNaLa [236], CodeSearchNet [237], Stack [238]

5.1 Parameter-Frozen Strategies

Current LLMs demonstrate increasingly greater generalization, zero-shot, and domain transfer capabilities, which enable them to deliver strong results across diverse task domains even with little fine-tuning. Parameter-frozen strategies directly integrate off-the-shelf retrievers, generators, and other modules, supplemented by training-free techniques, such as prompt engineering and the design of specific workflows [28, 103]. For example, PICa [28] adapts GPT-3 to solve VQA tasks by feeding it the retrieved VQA examples in a few-shot manner.

Parameter-frozen strategies substantially reduce the need for data annotation and model fine-tuning, thereby significantly lowering training costs. Moreover, the modular and decoupled design enables easy replacement or reuse of components to accommodate various scenarios. However, despite leveraging LLMs’ generalization ability, such strategies often lack fine-grained adaptability, making it difficult to fully optimize performance for specialized tasks or domains.

5.2 Parameter-Trainable Strategies

Parameter-trainable strategies train the LLM backbone and update its parameters to better align with specific tasks, especially in cases involving complex domain-specific knowledge, or when the input data shows a clear deviation from the patterns seen during pre-training. Despite higher training costs and computational demands, such strategies are extensively employed by researchers as they optimize the retriever and generator jointly or partly and can often achieve much better

performance compared to parameter-frozen strategies. Based on whether the optimization is restricted to isolated modules or performed jointly, parameter-trainable strategies can be further classified into *modular training* and *end-to-end training*.

5.2.1 Modular Training. The core characteristic of modular training is the independent optimization of individual components of MM-RAG, offering high flexibility. However, the absence of joint optimization may result in poor inter-module coordination. We discuss how different components are trained individually below.

Training of the Retriever only applies to dense retrievers, which is largely through contrastive learning. It typically employs a dual-tower architecture [15, 60, 239], utilizing two independent encoders to encode the query q and documents $Z = \{z_i\}$, both of which may be multimodal. The training usually uses loss functions like InfoNCE [240, 241] and Triplet loss [242]. In both scenarios, the optimization objective is maximizing the similarity $\text{sim}(q, z^+)$ between the query and the positive samples while minimizing $\text{sim}(q, z^-)$ between the query and the negative samples. Sparse retrievers are typically rule-based or statistical models and thus often do not require training.

Training of the Generator involves training the majority of parameters of an MM-RAG system. Although training the generator is resource-intensive, many researchers still opt to do it to get better performance. Rather than full parameter fine-tuning, Parameter-Efficient Fine-Tuning [243] methods such as LoRA [244] and p-tuning v2 [245] are commonly used to achieve good performance but with much lower fine-tuning costs. For example, RACC [31] converts compressed prompts into a p-tuning v2 modulation for the downstream MLLM to answer visual questions, while RA-VQAv2 [37] performs the same task by using LoRA to fine-tune BLIP2.

Adding Extra Parameter Layers refers to introducing functionality-specific intermediate parameter layers between the retriever and generator, which are trained separately from the main components. During the pretraining phase, the retriever and generator pursue different optimization objectives, which results in a semantic gap between the two components. On one hand, these additional layers can lessen the gap between the retriever and generator while improving the overall system performance. On the other hand, they are able to refine, re-encode, or re-weight the retrieved information, and make retrieved documents better utilized. For example, SmallCap[41] and RECAP[65] both bridge a frozen encoder (CLIP or CLAP) and a frozen GPT-2 decoder using multi-head cross-attention for image and audio captioning, respectively. EgoInstructor [77] inserts a perceiver resampler as a bottleneck module between the retriever and generator, which captures the relationships between the visual features in video inputs.

5.2.2 End-to-end Training. End-to-end training trains the retriever and the generator jointly, thus allowing for a holistic optimization of the system. For example, RA-VQA [25] and ReVeal [32] both jointly train the retriever and generator using a combined objective that includes a contrastive loss and a next-token prediction loss. Nonetheless, end-to-end training incurs substantial computational costs and involves complex optimization procedures. Furthermore, the low degree of modularity in end-to-end training may constrain deployment adaptability and complicate maintenance.

6 EVALUATION AND BENCHMARKS

This section discusses the evaluation metrics and benchmarks of the retriever and the generator as they are the two core components that affect the performance of MM-RAG. Figure 4 shows the metrics used for the evaluation of various stages of the system.

6.1 Evaluation of the Retriever

The retriever selects relevant documents from a knowledge base based on a given query. Therefore, the evaluation of the retriever involves both the relevance of retrieved documents to the query

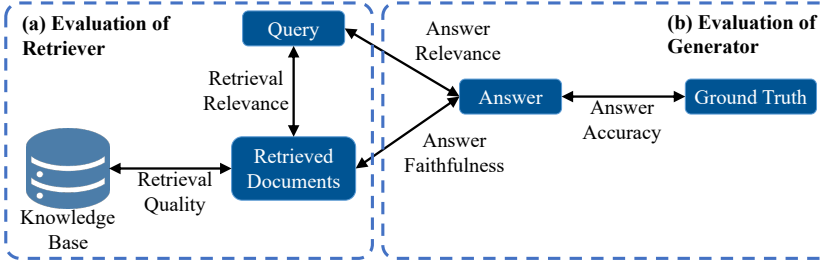


Fig. 4. Evaluation of Two Core Components in MM-RAG

and the quality of the retrieval relative to the candidate documents, which correspond to *Retrieval Relevance* and *Retrieval Quality*, respectively.

Retrieval Relevance (Retrieved Documents $\leftrightarrow$ Query) measures how well the retrieved documents address the information needs expressed in the query, ensuring the retrieval results are contextually aligned with the query’s intent and helpful in answering it. To quantify it, traditional approaches involve computing the semantic similarity between the retrieved documents and the query, using metrics such as cosine similarity and KL divergence of their dense embeddings. Lately, LLMs may be employed to directly assess whether the documents are helpful and contextually aligned with the query, providing a more flexible and human-like evaluation. For example, eRAG [246] uses the downstream LLM within the RAG system as a document annotator to assess retrieval quality.

Retrieval Quality (Retrieved Documents $\leftrightarrow$ Knowledge Base) is often quantified by traditional metrics from information retrieval and recommender systems such as Precision, Recall, F1-score, MRR, mAP, NDCG, and hit rate. It is particularly suitable when a set of golden documents—i.e., documents labeled as relevant—is available.

6.2 Evaluation of the Generator

The generator aims to produce outputs that answer the user’s query, and are consistent with the ground truth and the retrieved documents. The evaluation focuses on three key properties:

Answer Relevance (Answer $\leftrightarrow$ Query) evaluates how well the generated answer relates to the user’s query. High-quality answers effectively reflect the user’s needs and respond directly to the query.

Answer Correctness (Answer $\leftrightarrow$ Ground Truth) evaluates the correctness of the generated answer in comparison with the ground truth.

Answer Faithfulness (Answer $\leftrightarrow$ Retrieved Documents) examines if the generated answer is consistent with the retrieved documents.

The specific evaluation metrics depend on the downstream generation task and modality, as shown in Table 5. Most existing metrics reflect answer relevance and answer correctness. However, few metrics are explicitly designed to evaluate answer faithfulness, which remains an underexplored yet crucial direction. In addition to traditional metrics, LLM-as-a-judge [247] offers a promising direction for more nuanced evaluations, especially for multimodal tasks.

6.3 Benchmarks

The research on MM-RAG is still in its early stage. Although several benchmarks have been proposed, they predominantly focus on text and images, with most of them for VQA tasks, as summarized in Table 6. It is compelling to develop a more diverse and expansive set of evaluation benchmarks.

Table 5. Evaluation Metrics for Different Modality Generation Tasks

Modality Task	Evaluation Metric
Text generation	Accuracy, METEOR, CIDEr, SPICE, EM, BLEU, ROUGE/ROUGE-L, F1, BertScore [248], VQAScore, PPL, LLM as a judge [247]
Code generation	EM, Longest Common Subsequence, BLEU, CodeBLEU, ROUGE, Edit Distance, METEOR, CIDEr, Pass Rate, PPL
Image generation	SSIM, LPIPS, Inception Score, FID, CLIPScore [249]
Audio generation	APD, ASP, KL Divergence, Inception Score, FAD
Video generation	FVD, KVD
3D data generation	FID, Diversity, R-Precision, CLIPScore [249]

Table 6. Existing Benchmarks for MM-RAG

Modality	Benchmark	Evaluation Targets
Text + Image → Text	WebQA [210] OK-VQA [125] A-OKVQA [126]	Textual knowledge retrieval and reasoning.
	MRAG-BENCH [250] Visual-RAG [184]	Visual knowledge retrieval and reasoning.
	M ² RAG [251] MRAMG-Bench [252]	Document retrieval and generation with interleaved text-image content.
	Dyn-VQA [174]	Ability of adapting to rapidly changing multimodal knowledge; multi-hop and multimodal reasoning.
	CogBench [166]	Adaptive information acquisition by recording the entire planning procedure.
	Liu et al. [253]	Evaluates MM-RAG across four tasks: image captioning, multimodal QA, fact verification, and image reranking.
	OMG-QA [173] PDF-MVQA [254] Real-MM-RAG [255]	Retrieval and reasoning over complex document structures.
Text → Text	RAGAS [256] ARES [257]	Context relevance, answer faithfulness, and answer relevance.
	RGB [258]	Noise robustness, negative rejection, information integration, and counterfactual robustness.

Earlier benchmarks like WebQA [210], OK-VQA [125] and its successor A-OKVQA [126], which challenge models to retrieve outside open-domain knowledge and perform reasoning to answer visual questions, are most widely used. MRAG-BENCH [250] and Visual-RAG [184] both evaluate MM-RAG’s ability to retrieve and reason on visual knowledge. M²RAG [251] and MRAMG-Bench [252] each construct QA datasets containing multimodal data sources, which challenge models to retrieve and generate documents with interleaved text-image content, which requires models to determine not only whether an image should be included but also the appropriate position for its insertion. Dyn-VQA [174] introduces dynamic visual questions, including those requiring rapidly changing multimodal knowledge and multi-hop reasoning. These necessitate that the model dynamically adjusts its choice of tools and timing for retrieval and generation. CogBench [166] provides a comprehensive record of the entire planning procedure involved in multimodal question answering, thereby evaluating MLLMs’ ability of adaptive information acquisition and reasoning.

Liu et al. [253] propose a benchmark that evaluates the ability of MM-RAG through four tasks: image captioning, multimodal question answering, multimodal fact verification, and image reranking. OMG-QA [173], PDF-MVQA [254], and Real-MM-RAG [255] evaluate MM-RAG systems' ability to retrieve and reason across visually-rich documents, including text, tables, and images, to generate answers. The latter introduces a query rewriting step that alters the phrasing and structure of questions, making the retrieval process more challenging.

Traditional text-only RAG benchmarks can still provide a foundation for evaluating and understanding MM-RAG system performance. ARES [257] and RAGAS [256] let LLM evaluate query-passage-answer triples from three perspectives: context relevance, answer faithfulness, and answer relevance through few-shot learning and prompt engineering, respectively. By replacing LLM with MLLM, this pipeline can be applied to the evaluation of MM-RAG. RGB [258] evaluates RAG capabilities along four dimensions: noise robustness, negative rejection, information integration, and counterfactual robustness. Though text-based, its evaluation dimensions are equally applicable to MM-RAG systems.

7 APPLICATIONS

MM-RAG significantly enhances the contextual comprehension and generative capabilities of AI systems, and thus demonstrates great application prospects, as discussed below.

7.1 Healthcare and Medicine

The core applications of MM-RAG in the healthcare domain include medical imaging analysis, clinical decision support, and drug discovery. DeltaNet [46] retrieves medical images and reports from medical records based on current medical images and conditionally generates the final report. MMed-RAG [38] and FactMM-RAG [48] introduce RAG to enhance the factuality of Med-LVLMs on report generation tasks involving radiology, ophthalmology, and pathology. IRDIFF [183], a retrieval-augmented 3D diffusion model, generates realistic molecules with strong binding affinities and desirable properties. EMERGE [178] utilizes time-series signals and clinical notes from multimodal electronic health records as a knowledge base to support predictions of patient death during hospitalization and 30-day readmission likelihood.

7.2 Finance

Leveraging financial market-specific data (e.g., K-line charts, financial statements, contracts, and bills), MM-RAG can establish knowledge repositories, with which it supports interpretable investment and risk management solutions [259, 260]. Furthermore, by leveraging time-series data of stock price trends and related financial indicators, MM-RAG has even been extended to stock movement prediction applications [139, 261].

7.3 Traffic

MM-RAG shows great promise in traffic scenarios, particularly in complex tasks requiring the integration of visual perception and knowledge-based reasoning. RealGen [262] generates new scenarios for autonomous driving training by combining behaviors from multiple retrieved examples. RAG-Driver [142] produces driving action explanations, justifications, and control signal predictions by retrieving expert demonstrations.

7.4 Education

MM-RAG can integrate multimodal data from educational scenarios (e.g., textbook documents, exercise databases, instructional videos, and presentation charts). VL-RAG [263] customizes answer

and image databases for teaching scenarios to retrieve relevant information during the teaching process. This facilitates automated student teaching with minimal human oversight, while preserving scalability across diverse subjects and curricular materials.

8 DISCUSSION AND FUTURE DIRECTIONS

8.1 Unexplored Input-Output Combinations

As shown in Table 1, existing MM-RAG studies have focused on only a few input-output modality combinations, and many remain underexplored. Each of them may correspond to an important and exciting group of applications. Systematic exploration of these combinations may lead to novel research directions.

8.2 Limited Cross-Modal Alignment

Existing multimodal models have shown relatively strong performance on well-studied modality combinations such as text-image. However, the performance remains poor for complex or understudied modality pairs (e.g., 3D data and video) and need further investigation.

8.3 Robustness and Safety

The integration of multimodal queries and knowledge in MM-RAG consequently endows its robustness and safety research with heightened complexity and urgency. Noise [198, 264] in multimodal documents (e.g. misleading images or deepfake videos) and knowledge conflicts [264–266] arising from disparate modalities and data sources may severely degrade both retrieval and generation quality. Concurrently, ambiguous user queries in a multimodal interactive setting, corpus poisoning attacks, and prompt-based adversarial exploits targeting the retrieval or generation stages may amplify system vulnerabilities at every stage [267]. Although preliminary defense strategies have been proposed for text-based RAG [268, 269], the inherent heterogeneity of MM-RAG renders these approaches insufficient and requires further research.

8.4 Adapting Text-based RAG Methods to Multimodal Settings

Many text-based RAG methods have been developed in recent years. We may transfer successful ideas from text-only RAG to MM-RAG. Recent developments such as Modular RAG [270], Parametric RAG [271], and Agentic RAG [272] have demonstrated the potential of more structured, flexible, and intelligent RAG architectures in the textual domain. These methods offer valuable insights that could inform the design of more capable MM-RAG systems. By exploring how these methods can be generalized in multimodal settings, future research can help bridge the gap between textual and multimodal RAG, ultimately enabling more powerful and general-purpose AI systems.

8.5 Limited Benchmarks

Comprehensive investigation into the evaluation frameworks of MM-RAG remains scarce, especially for modalities beyond text and images. A unified and comprehensive benchmark is a great challenge and void in MM-RAG research. Addressing this issue requires collaborative efforts from the research community to curate robust and standardized datasets, formulate multidimensional evaluation metrics, and foster collaboration and healthy competition.

8.6 Relationship with Other Technologies

Although research on MM-RAG remains sparse, its potential can be greatly expanded by integrating with complementary technologies such as personalization [273–275], agents [272], and embodied intelligence [276, 277], enabling user-tailored responses, richer human-agent collaboration, and

grounding in real-world perception and action. At the same time, MM-RAG can benefit from the methodological legacy of expert systems [278], whose paradigms of knowledge acquisition, validation, and user interaction remain highly relevant across domains. Together, these connections not only mitigate the limitations of standalone approaches but also provide practical guidance for the industrial adoption and domain-specific application of MM-RAG.

9 CONCLUSIONS

In this paper, we present a comprehensive survey of the recent advances in MM-RAG methods, structured around the many combinations of input and output modalities. Specifically, we introduce a taxonomy of MM-RAG methods based on their input-output modality combinations. Based on the taxonomy, we carry out a detailed examination of the essential stages of MM-RAG systems and their training strategies. Despite its promising potential and wide range of future applications, research in this domain is in its early stage and faces challenges that demand sustained efforts.

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